

# GNU-Plot-Lab-Notebook

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## Section 1

### Question 1

Plot the following functions for  $-10 \leq x \leq 10$  on the same graph using GNUPLOT. Set 1000 samples. Take the range of  $y$ -values from  $-4$  to  $+4$ . Mark each curve with different line styles and line widths. Label  $x$ -axis and  $y$ -axis as ' $x$ -values' and ' $y$ -values' respectively. Set the title of the plot as 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1  reset
2  set terminal svg background "white"
3  set output "Lab-files/Output/1/Q_xxi_ou.svg"
4
5  set title "Function plots"
6  set grid
7  set xzeroaxis linestyle 1 lc "black"
8  set yzeroaxis linestyle 1 lc "black"
9  set xrange [-10:10]
10 set yrange [-4:4]
11 set xlabel "x-values"
12 set ylabel "y-values"
13 set samples 1000
14
15 f1(x)=(1/(tanh(x)))-(1/x)
16 f2(x)=exp(-sin(x))
17 f3(x)=x*abs(x)
18
19 plot f1(x) lw 1 lc "#5f0f40" dt 1, f2(x) lw 2 lc "#fb8b24" dt 2, f3(x) lw 3 lc "#9
    a031e" dt 3
```

The output plot is given in the following page.

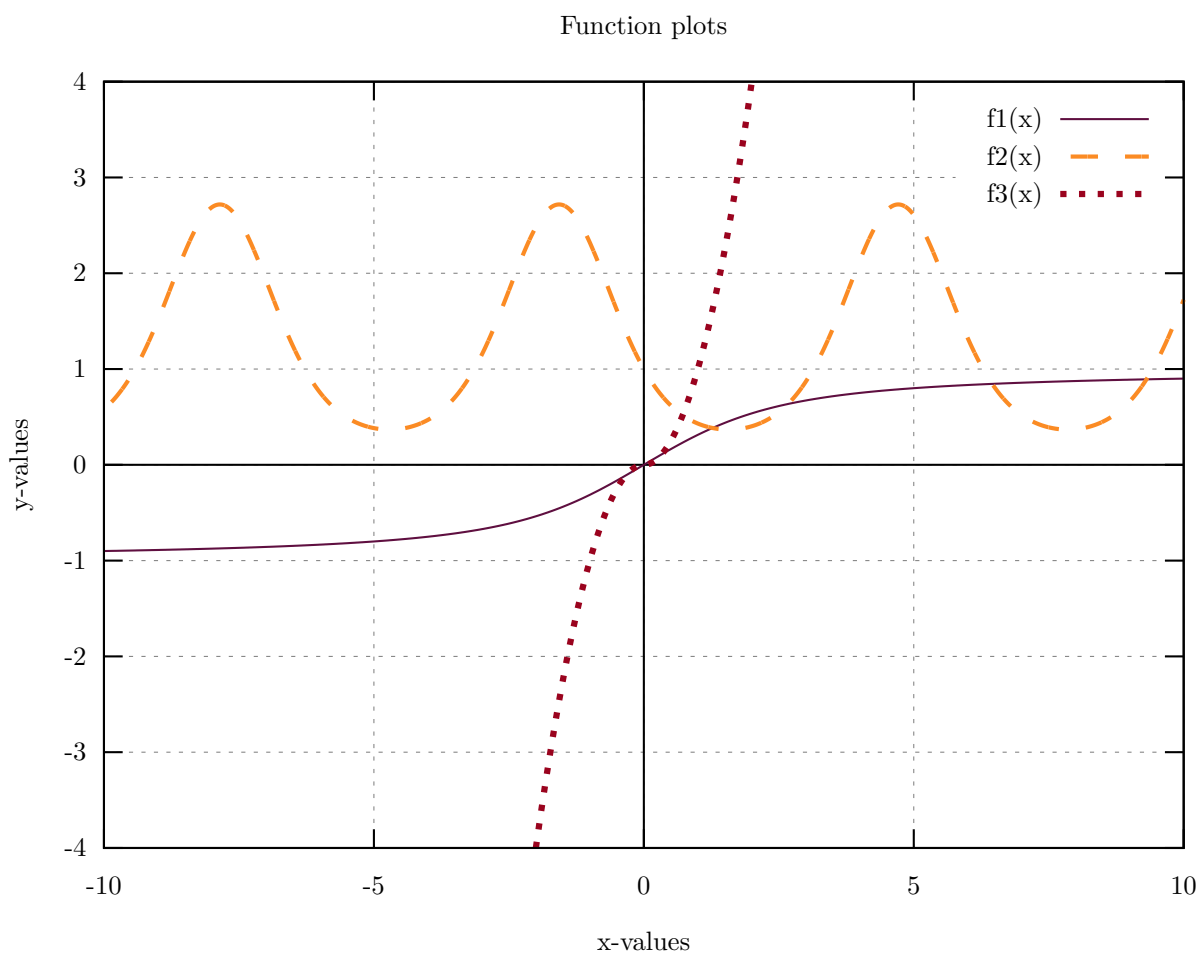


Figure 1:  $f_1(x) = \coth(x) - \frac{1}{x}$ ,  $f_2(x) = e^{-\sin(x)}$ ,  $f_3(x) = x|x|$

## Question 2

Plot the following functions for  $0 \leq x \leq \pi$  on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \sin^2(\sqrt{x}), \quad f_2(x) = e^{-x/2}, \quad f_3(x) = 12.34x^{5/3}$$

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xx_ou.svg"
4
5 set title "Function plots"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [0:pi]
10 set yrange [0:8]
11 set xtics 0.5
12 set ytics 1
13 set xlabel "x-values"
14 set ylabel "y-values"
15
16 f1(x)=sin(sqrt(x))*2
17 f2(x)=exp(-x/2)
18 f3(x)=12.34*(x**(5/3))
19
20 set samples 1000
21 plot f1(x) lw 1 dt 1 lc "#780000", f2(x) lw 2 dt 2 lc "#c1121f", f3(x) lw 3 dt 3
    lc "#003049"
```

The output plot is given in the following page.

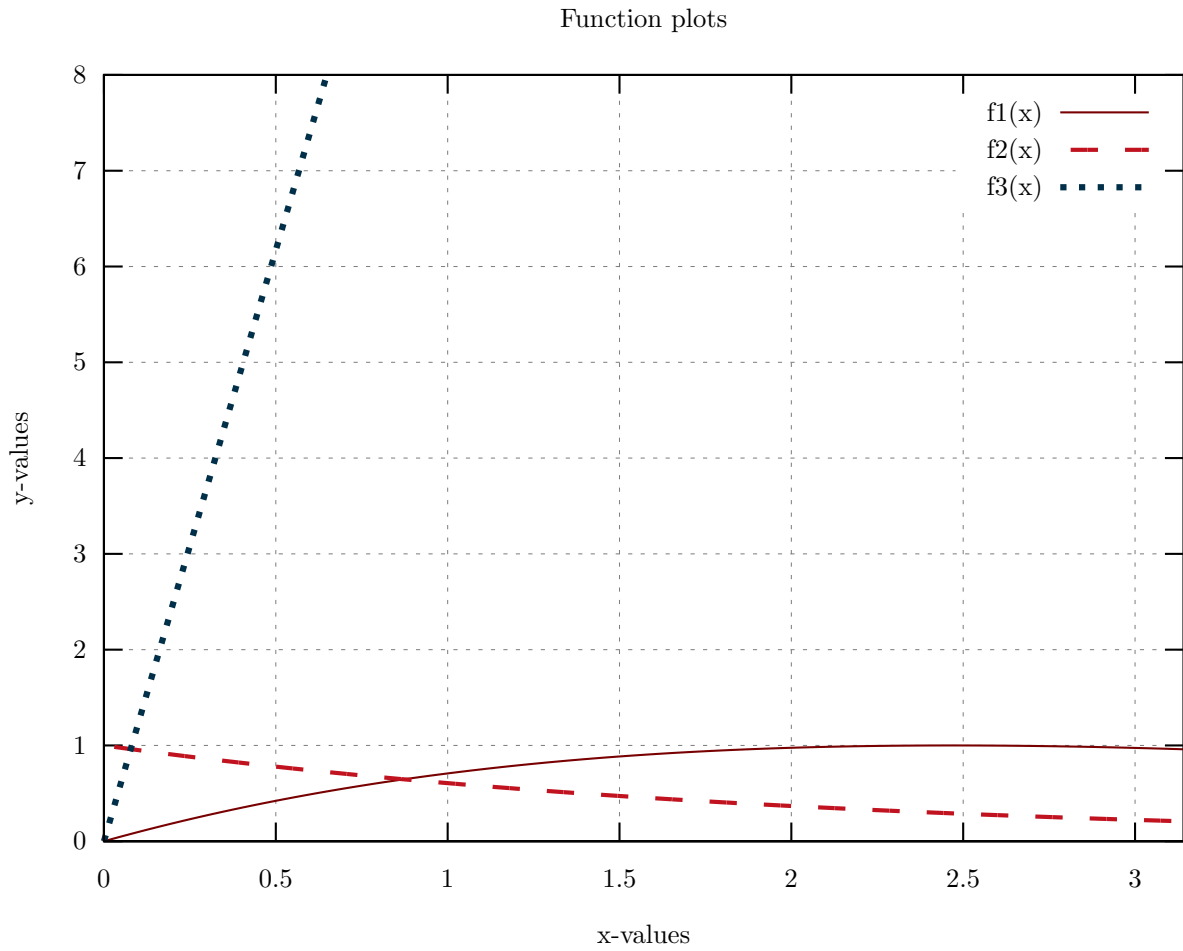


Figure 2:  $f_1(x) = \sin^2(\sqrt{x})$  ,  $f_2(x) = e^{-x/2}$  ,  $f_3(x) = 12.34x^{5/3}$

### Question 3

Define three functions in the given range:

$$\begin{aligned} f_1(x) &= \ln\left(\frac{1}{x}\right), & x > 0.1, \\ f_2(x) &= 3x^2 \sin x, & -1 < x < 0.1, \\ f_3(x) &= \cos 3x, & x < -1. \end{aligned}$$

Set the range of  $x$  values from  $-3$  to  $+3$  and the range of  $y$  values from  $-2.5$  to  $+2.5$ . Show both the axes on the plot. Set  $y$  label "Piecewise continuous function". Set ticks of the  $y$  axis at an interval of  $0.5$ . Plot the three functions and store the plot in a file and print it.

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xiv_ou.svg"
4
5 set grid
6
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9
10 set ytics 0.5
11
12 set xrange [-3:3]
13 set yrange [-2.5:2.5]
14
15 set ylabel "Piecewise continuous function"
16
17
18 f1(x)=(x>0.1)?(log(1/x)):NaN #Not as Number
19 f2(x)=(x<0.1 && x>-1)?(3*(x**2)*sin(x)):NaN
20 f3(x)=(x<-1)?(cos(3*x)):NaN
21
22 plot f1(x) lc "#9b2226" lw 2 , f2(x) lc "#bb3e03" lw 2 , f3(x) lc "#ee9b00" lw 2
```

The output plot is given in the figure below.

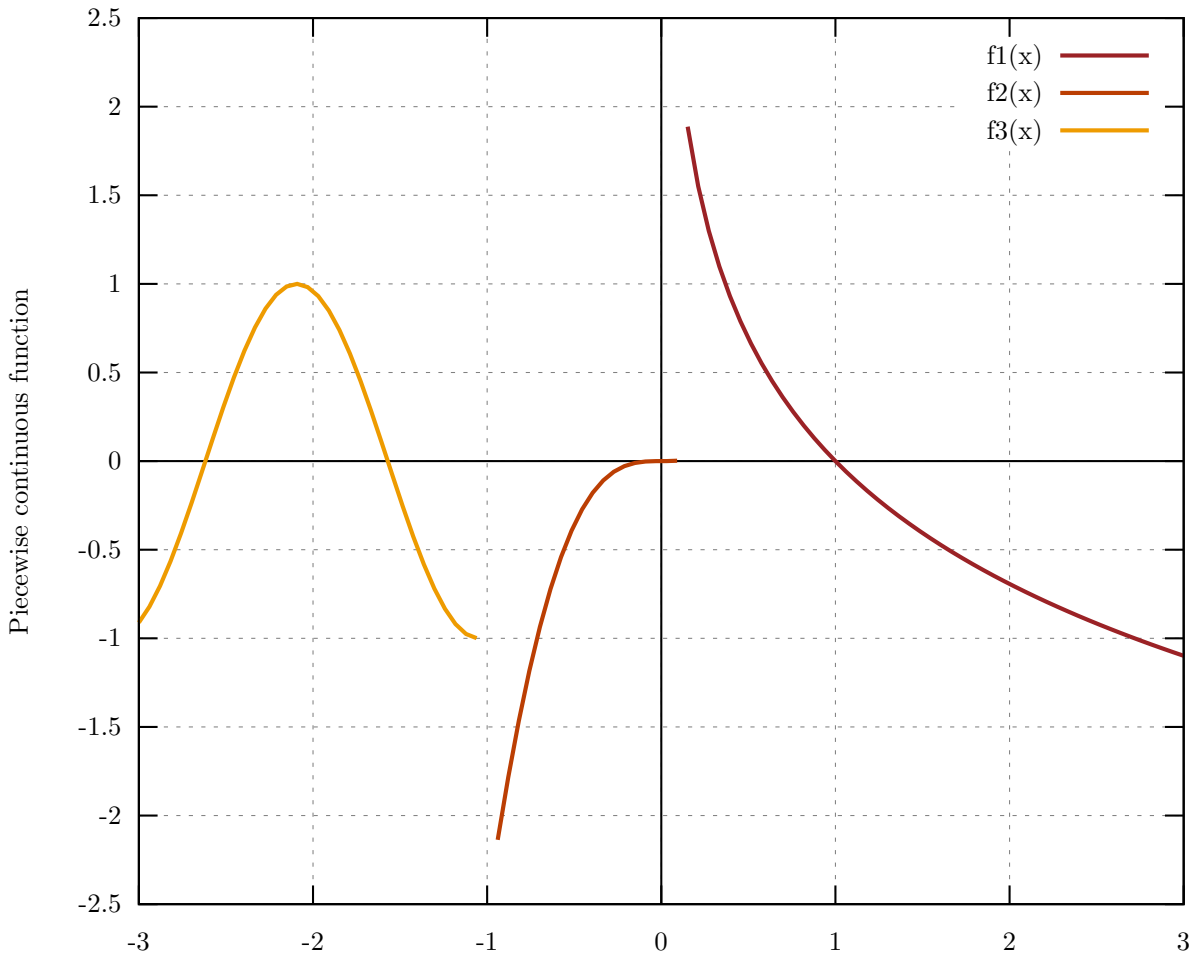


Figure 3: Plot of the piecewise continuous functions  $f_1(x)$ ,  $f_2(x)$ , and  $f_3(x)$

## Question 4

Plot

$$2x \sin 2x, \quad \frac{1}{2}x \sinh\left(\frac{x}{2}\right)$$

on the same graph. The line width of the two curves to be different. Set the range of  $x$  values from  $-\pi$  to  $+\pi$  and the range of  $y$  values from  $-5$  to  $5$ . Show both the axes on the plot. Ticks of  $x$  axis will be at interval of  $0.5$ . The label of  $x$  and  $y$  axis to be "x values" and "y values" respectively. Store the plot in a file and print it.

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1  reset
2  set terminal svg background "white"
3  set output "Lab-files/Output/1/Q_xvi_ou.svg"
4
5  set grid
6  set xzeroaxis linestyle 1 lc "black"
7  set yzeroaxis linestyle 1 lc "black"
8  set xrange [-pi:pi]
9  set yrange [-5:5]
10 set xtics 0.5
11 set xlabel "x values"
12 set ylabel "y values"
13
14 f1(x)=2*x*sin(2*x)
15 f2(x)=(x/2)*sinh(x/2)
16
17 plot f1(x) lc "#9a031e" lw 2, f2(x) lc "#0f4c5c" lw 3
```

The output plot is given in the figure below.

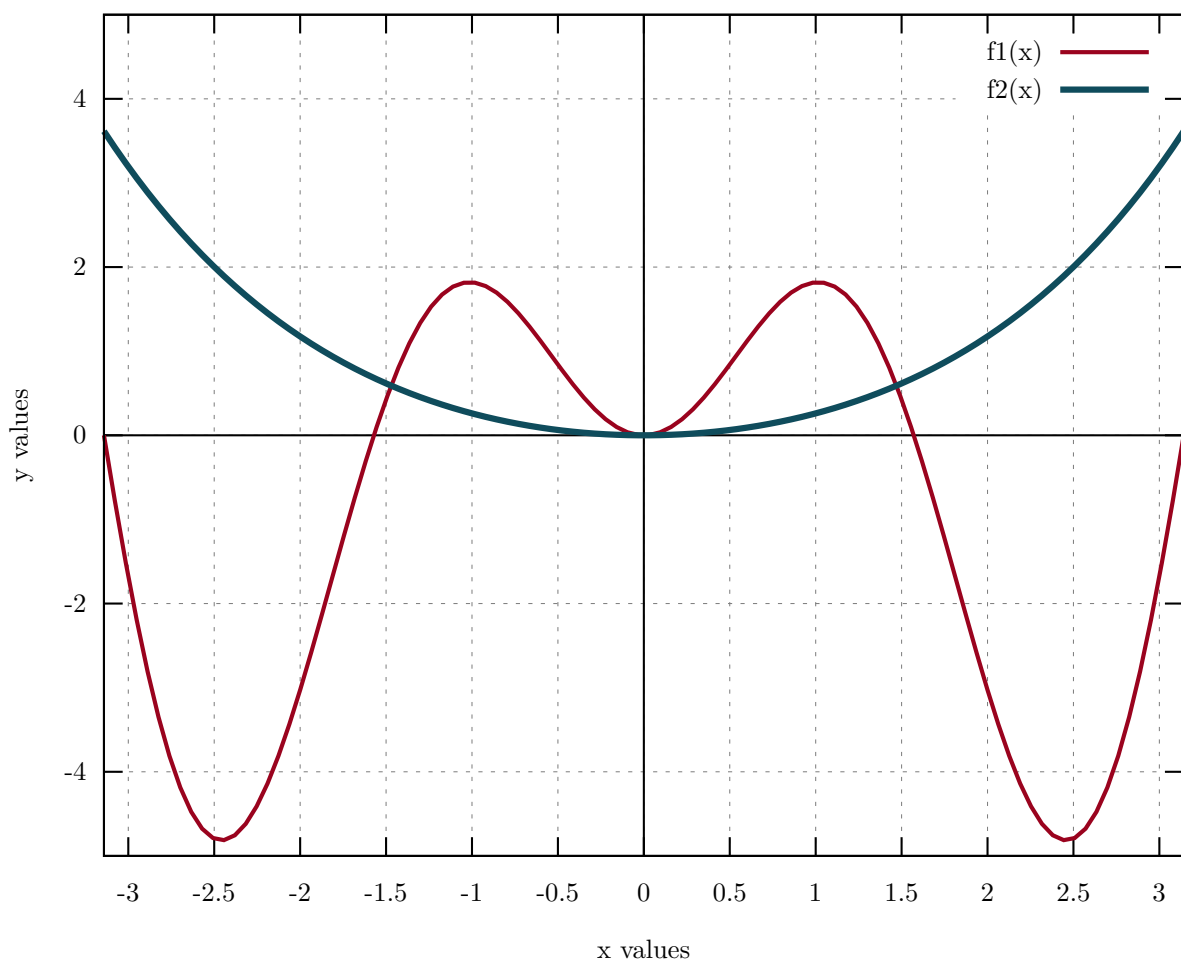


Figure 4:  $2x \sin 2x$ ,  $\frac{1}{2}x \sinh\left(\frac{x}{2}\right)$

## Question 5

**Problem Set 1 (iii).** Plot  $\sin(x)$  and  $2\sin(-x)$  simultaneously on the same plot within the range  $-3\pi$  to  $3\pi$ . Use the system-defined value of  $3\pi$  as `3*pi`. Label the  $x$ -axis as “x values” and the  $y$ -axis as “standing wave”. Save the output plot in a file.

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_iii_ou.svg"
4
5 set grid
6 set xzeroaxis linetype 1 lc "black"
7 set yzeroaxis linetype 1 lc "black"
8 set xrange [-3*pi:3*pi]
9 set yrange [-3:3]
10 set xlabel "x values"
11 set ylabel "standing wave"
12
13 plot sin(x) lc "orchid4" lw 2, 2*sin(-x) lw 2
```

The output plot is given in the figure below.

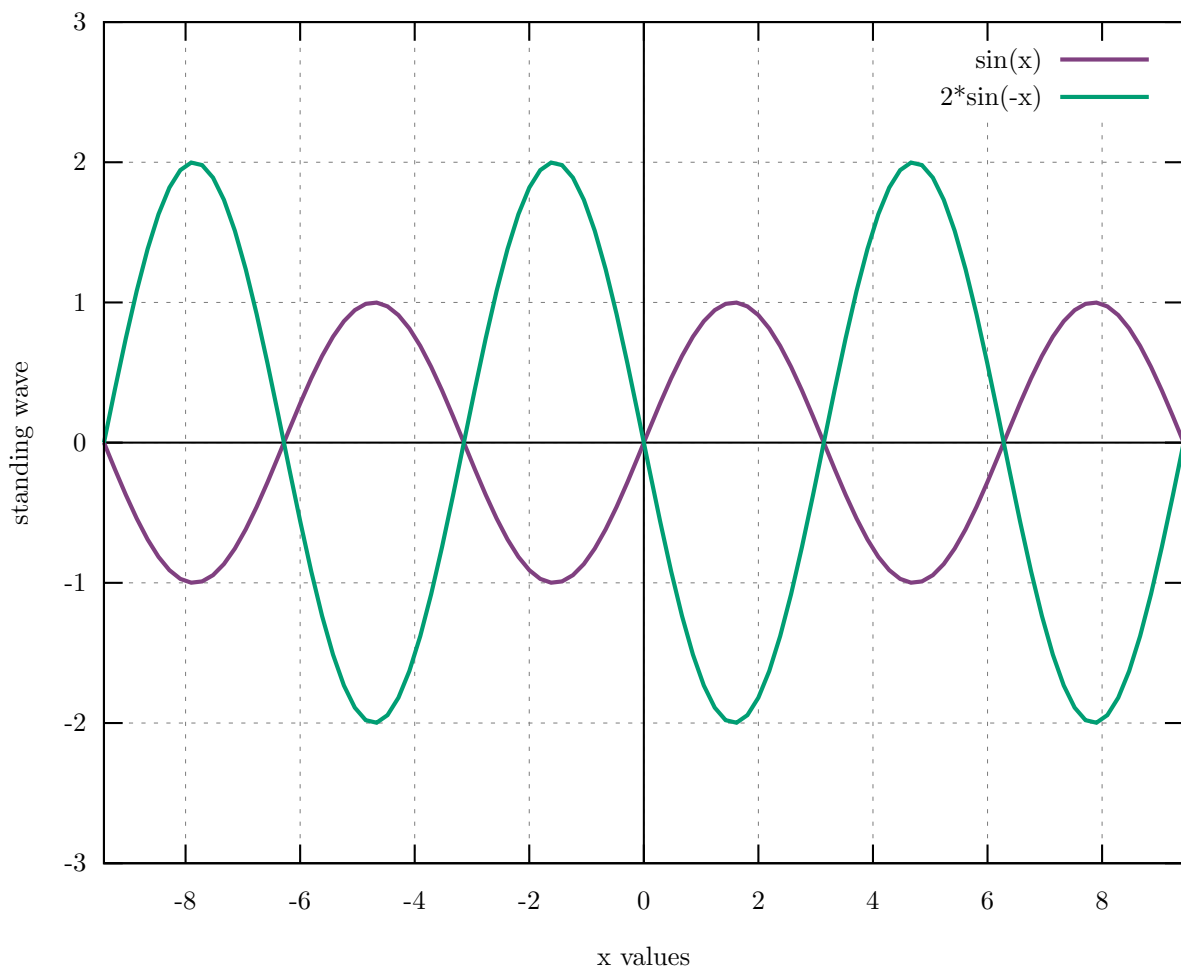


Figure 5: The functions  $\sin(x)$  and  $2\sin(-x)$  plotted over  $-3\pi \leq x \leq 3\pi$ .

## Question 6

**Problem Set 1 (xiii).** Define the following piecewise function:

$$f(x) = \begin{cases} x \cos x, & x < 0, \\ x^2 + 3x, & 0 \leq x \leq 1, \\ \ln x + 4, & x > 1. \end{cases}$$

Plot it for  $-3 \leq x \leq 3$  and  $-2 \leq y \leq 5$ . Show both axes and set the title to “Continuous function without derivative at all points”. Save the plot in a file.

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xiii_ou.svg"
4
5 set title "Continuous function without derivative at all points"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [-3:3]
10 set yrange [-2:5]
11
12 f(x)=(x<0)?(x*cos(x)):(x>=0 && x<=1)?(x**2 + 3*x):(x>1)?(log(x)+4):NaN
13
14 plot f(x) lc "#ee6c4d" lw 2
```

The output plot is given in the figure below.

Continuous function without derivative at all points

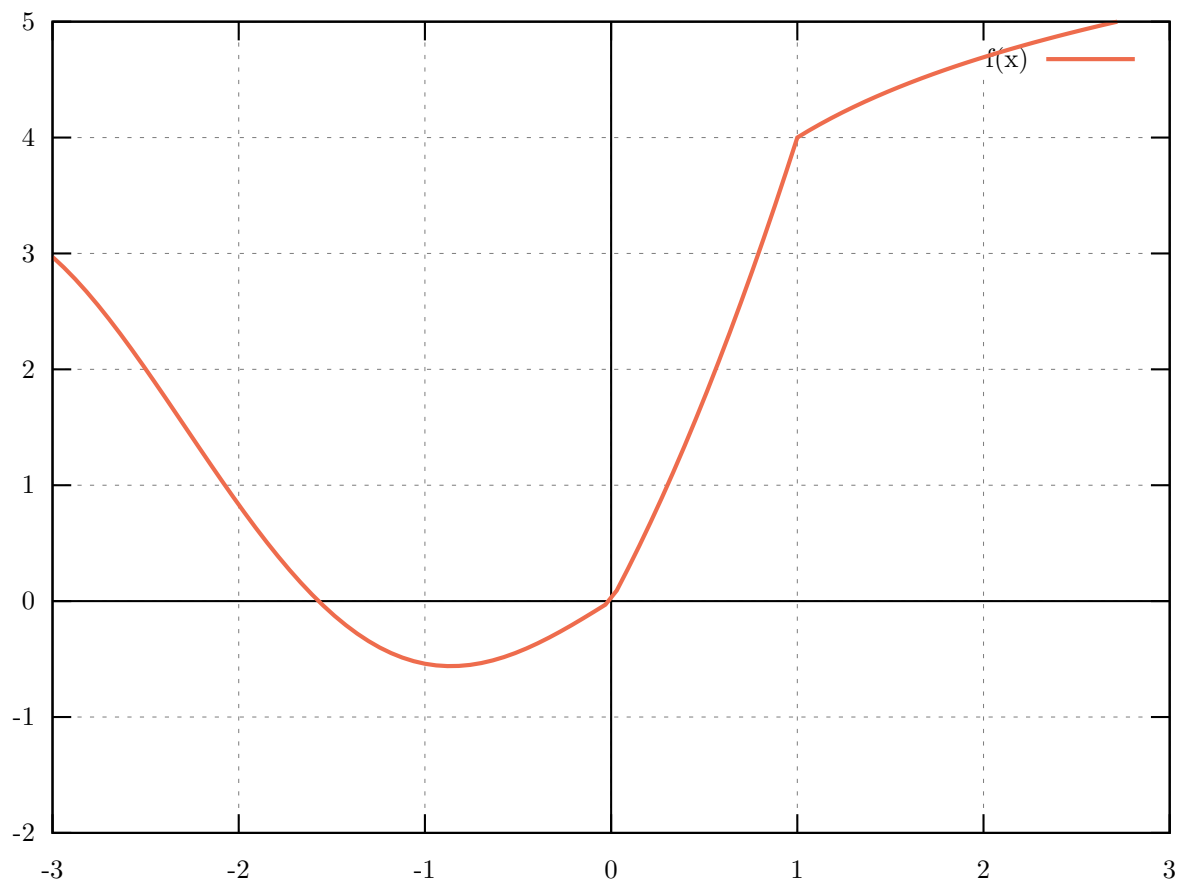


Figure 6: A continuous piecewise function with changes in derivative.

## Question 7

**Problem Set 1 (xii).** Plot  $\frac{\sin x}{x}$ ,  $\frac{\sinh x}{x}$ , and  $y = 1$  on the same graph. Set the  $x$ -range from  $-\pi$  to  $+\pi$  and the  $y$ -range from 0 to 5. Show the  $y$ -axis, set the title to “Plot of different functions of  $x$ ”, and label the axes “ $x$  values” and “ $y$  values”. Use different line widths for all three curves and save the plot in a file.

**Sol:** The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xii_ou.svg"
4
5 set grid
6 set title "Plot of different functions of x"
7 set xrange [-pi:pi]
8 set yrange [0:5]
9 set xtics 0.5
10 set ytics 1
11 set xlabel "x values"
12 set ylabel "y values"
13 set xzeroaxis linestyle 1 lc "black"
14 set yzeroaxis linestyle 1 lc "black"
15
16 y1(x)=(x==0)?1:sin(x)/x
17 y2(x)=(x==0)?1:sinh(x)/x
18 y3(x)=1
19
20 plot y1(x) lc "#6F1D1B" lw 1, y2(x) lc "#fb8500" lw 2, y3(x) lc "#f94144" lw 3
```

The output plot is given in the figure below.

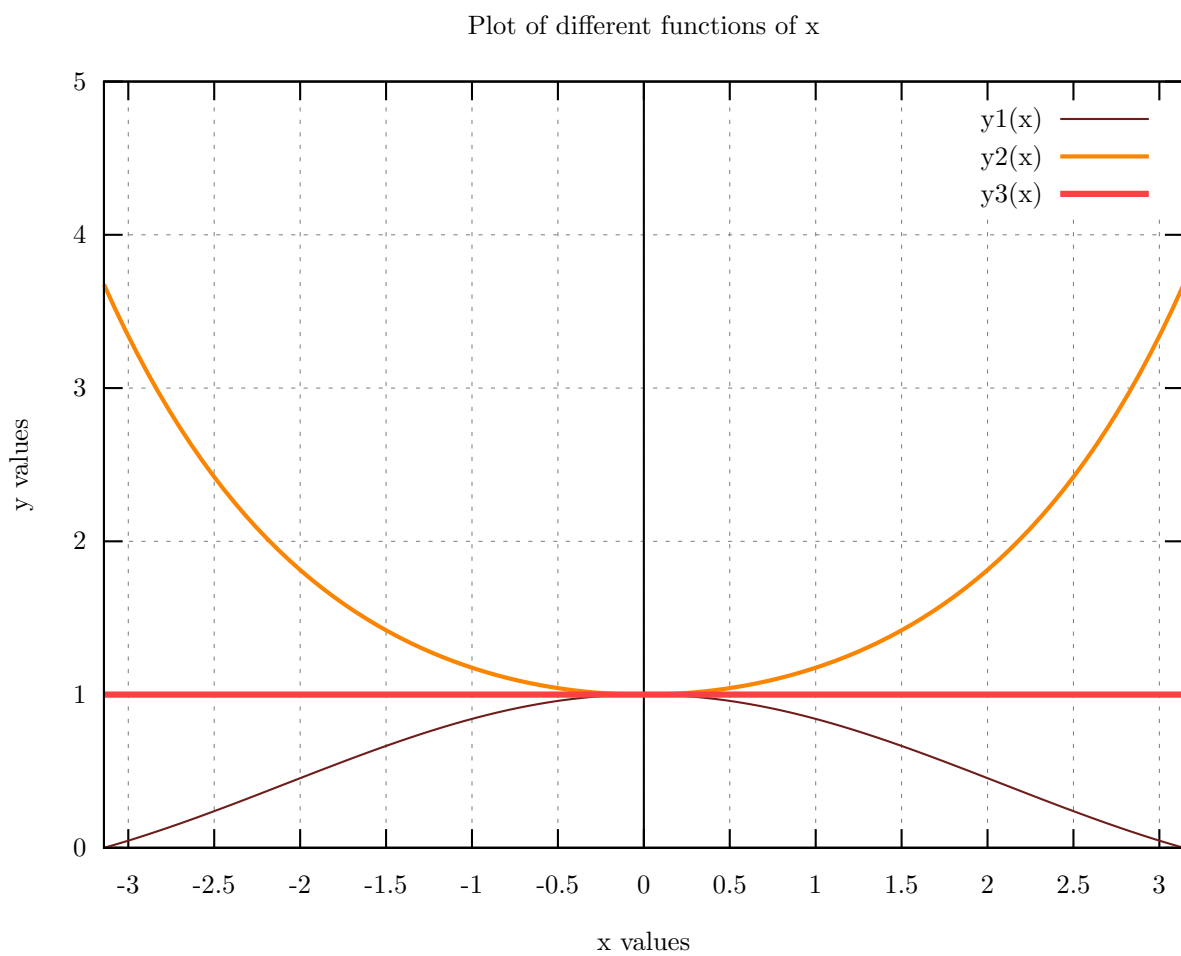


Figure 7: The functions  $\frac{\sin x}{x}$ ,  $\frac{\sinh x}{x}$ , and  $y = 1$ .